

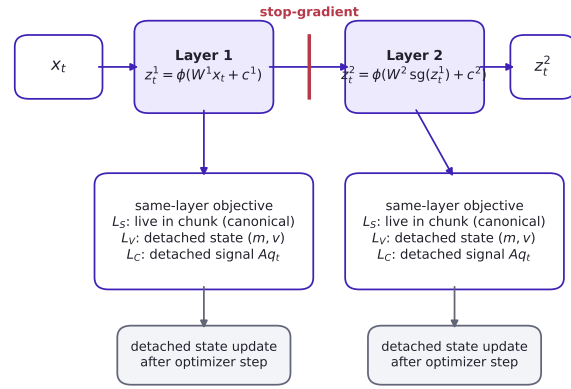
Temporal Regularized Learning: Competitive Deep Local Soft-SFA

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A Local credit assignment

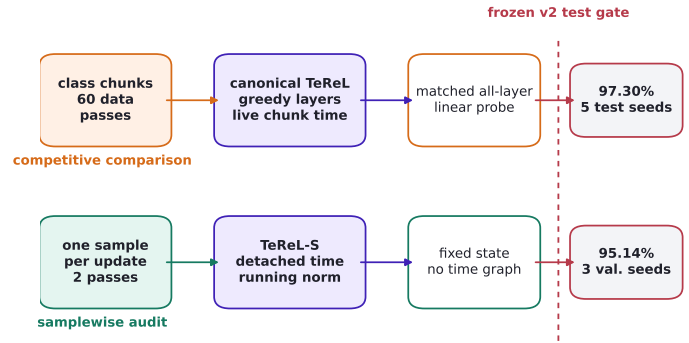
Every loss updates one nonlinear layer; graph cuts are explicit.



TeReL-S detaches p_t : $D^2 + 4D + 1$ state elements; no temporal graph is retained.

B Two locality regimes, retained accuracy

Matched readouts separate the competitive result from the streaming audit.



Selection: validation only (101/202/303) • Confirmation: frozen test matrix (1101-1505)

Figure 1: A: Each TEREL layer receives temporal-coherence, gated-variance, and lateral-decorrelation signals without an error path from later layers. **B:** Canonical TEREL retains short within-chunk temporal gradients; TEREL-S detaches them and admits bounded-state samplewise execution. Corrected experiments use matched probes, explicit compute accounting, and a manifest-locked test gate.

Abstract

Temporal coherence is an attractive learning signal when credit assignment must remain within a layer, but slowness alone admits a collapsed solution. We introduce Temporal Regularized Learning (TEREL), a deep local rule that combines temporal coherence, gated variance expansion, and a same-layer decorrelation signal. The objective is a soft neural form of Slow Feature Analysis (SFA), viewed through the variance–invariance–covariance structure of modern regularized self-supervision. We define two roles in the method family: canonical TEREL is greedy and layer-local while retaining gradients between adjacent activations inside a short chunk; TEREL-S detaches the temporal reference and supports bounded-state samplewise execution. We derive both gradients, specify update order and boundary handling, and account separately for parameters, optimizer state, dynamic state, and batch storage.

All headline experiments are rerun from an optimizer-corrected implementation under a validation-selected, manifest-locked protocol. On label-ordered MNIST, canonical TEREL reaches $97.30 \pm 0.07\%$ test accuracy across five seeds, versus $98.34 \pm 0.08\%$ for compute-matched backpropagation and $95.13 \pm 0.27\%$ for matched random features. Restricting the readout to the last layer retains $97.14 \pm 0.10\%$, and TEREL-S reaches $96.29 \pm 0.07\%$. Validation controls isolate the mechanism: removing temporal coherence loses 8.18 points, removing variance

expansion collapses feature scale, and removing decorrelation collapses effective rank. A separately tuned, strictly samplewise TEREL-S configuration reaches $95.14 \pm 0.04\%$ validation accuracy with no retained temporal graph and fixed state. A chronological PAMAP2 stress test remains inconclusive and is reported as a secondary boundary, not the paper’s center. These results restore the competitive empirical finding while making precise which locality, streaming, and natural-order conclusions the evidence supports.

Code: <https://github.com/Axym-Labs/TeReL>

Project page: <https://axym.org/work/terel>

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1 Introduction

Many observations change faster than the latent causes that generate them. Object identity persists while viewpoint and illumination vary; a person’s activity persists across noisy wearable-sensor readings; and the state of a dynamical system generally evolves more slowly than its measurement noise. Temporal proximity can therefore replace hand-designed paired views as a learning signal. The difficulty is that making adjacent representations equal also makes a constant representation optimal.

Classical Slow Feature Analysis (SFA) resolves this ambiguity by minimizing temporal derivatives subject to zero-mean, unit-variance, and decorrelation constraints [1]. Modern regularized self-supervised objectives arrive at a closely related structure: an invariance term is paired with variance and covariance regularizers to prevent collapse [2, 3]. The shared structure is not itself new. Our question is instead an implementation question: can this soft-SFA objective train each nonlinear layer without gradients from later layers or through temporal history?

TEREL answers with three same-layer signals. A temporal term attracts the current activation to its preceding activation. A neuron-local term expands activity while its detached running variance lies below a target. A dense same-layer operator tracks off-diagonal second moments and supplies a detached approximation to the covariance-penalty gradient. Detaching the input to every subsequent layer makes the parameter update spatially local in the credit-assignment sense. Canonical TEREL retains the preceding activation in a short within-chunk graph and is therefore layer-local but not strictly online. TEREL-S detaches that reference, making the rule causal and bounded-history. Minibatches may still execute either variant efficiently; batching is not itself the locality claim.

This distinction matters because three claims are easily conflated. First, local credit assignment does not imply sparse communication: the lateral operator communicates within a layer. Second, bounded history does not imply constant state in width: a dense width- D lateral operator costs D^2 . Third, an objective without label terms is not self-supervised if labels were used to construct its temporal order. We therefore use label-ordered MNIST as a controlled temporal-supervision study. Chronological wearable streams whose labels are excluded from encoder batches provide a secondary natural-order stress test; their inconclusive outcome is not allowed to displace the controlled competitive result.

The paper makes five contributions:

1. We formulate TEREL as a deep, layer-local soft-SFA rule and locate its contribution relative to SFA, Incremental SFA, Bio-SFA, VICReg, and blockwise self-supervised learning.
2. We give the implemented objective, exact activation and weight gradients for detached and undetached time, the detached state updates, and a quantitative error decomposition for the shifted lateral heuristic.
3. We recover competitive performance from an optimizer-corrected implementation: canonical TEREL lies 1.04 percentage points behind compute-matched backpropagation and 2.17 points

above matched random features, while a last-layer-only audit loses only 0.16 points. One-factor controls separately expose loss of temporal supervision, scale collapse, and redundancy collapse.

4. We demonstrate tuned batch-size-one learning with fixed state and no temporal graph, and report parameters, optimizer state, dynamic state, peak memory, compute proxies, and throughput separately.
5. We provide validation-only selection, matched architectures and readouts, five confirmatory seeds, raw results, uncertainty, and exact result-to-commit provenance.

2 Temporal Regularized Learning

2.1 From constrained SFA to a soft local objective

Let x_t be a temporal stream and let layer ℓ produce

$$a_t^\ell = W^\ell z_t^{\ell-1} + c^\ell, \quad z_t^\ell = \phi(a_t^\ell). \quad (1)$$

SFA searches for features that change slowly while excluding constant and duplicated solutions. In one common statement it minimizes $\mathbb{E}\|z_t - z_{t-1}\|_2^2$ subject to zero mean, unit variance, and decorrelated outputs [1]. TEREL replaces hard population constraints by detached running state and local soft terms. This is a particular neural parameterization and optimizer, not a new discovery of the SFA objective.

For one layer, consider B activations $z_t \in \mathbb{R}^D$. A Boolean boundary b_t marks the start of a subject, recording, or constructed chunk. Let $I_t = 1 - b_t$, additionally set to zero when no stored predecessor exists, and let $N_I = \sum_t I_t$. The state available before the step is a detached mean m , variance v , and lateral operator A . The canonical local loss is

$$L_S = \frac{1}{N_I D} \sum_{t: I_t=1} \|z_t - p_t\|_2^2, \quad (2)$$

$$L_V = -\frac{1}{BD} \sum_{t=1}^B \sum_{j=1}^D [\gamma - v_j]_+ (z_{tj} - m_j)^2, \quad (3)$$

$$L_C = \frac{1}{BD} \sum_{t=1}^B (z_t - m)^\top A q_t, \quad (4)$$

$$L = \lambda_S L_S + \lambda_V L_V + \lambda_C L_C. \quad (5)$$

If $N_I = 0$, L_S and its gradient are defined as zero. In canonical TEREL, $p_t = z_{t-1}$ inside a short chunk and $q_t = \text{sg}(z_t - m)$. The input to each subsequent layer remains detached, so these within-chunk temporal gradients never become gradients across depth. In TEREL-S, $p_t = \text{sg}(z_{t-1})$ (or the detached last activation from the preceding call). Boundary flags invalidate temporal pairs in both cases.

The minus sign in Eq. (3) is essential. With the gate $g_j = [\gamma - v_j]_+$ treated as state, minimizing L_V pushes a currently active unit away from its mean until its running variance reaches γ . The term is a local gradient surrogate for a variance constraint; it is not the usual nonnegative VICReg scalar variance penalty. Its effectiveness depends on the statistics timescale and normalization. We therefore treat state rates, normalization, and activation choice as algorithmic hyperparameters and select them on validation data.

2.2 Lateral decorrelation as a gradient proxy

Define the detached off-diagonal second-moment operator

$$C = \text{offdiag} \left(\frac{1}{B} \sum_t (z_t - m)(z_t - m)^\top \right). \quad (6)$$

The direct control used in our ablations minimizes $L_{\text{direct}} = \|C\|_F^2/D$. With m detached, its activation gradient is $4C(z_t - m)/(BD)$. TEREL instead tracks C with A and uses Eq. (4); because q_t and A are detached, the encoder receives $Aq_t/(BD)$. If $A \approx C$ and $q_t = z_t - m$, this matches the direct gradient direction up to a positive factor four. It is a proxy for the gradient, not an equality between scalar objectives.

The shifted heuristic uses the preceding centered activation as q_t . Let $u_t = z_t - m$. Its error relative to the direct signal obeys

$$\|Aq_t - Cu_t\|_2 \leq \|A - C\|_2 \|u_t\|_2 + \|A\|_2 \|q_t - u_t\|_2. \quad (7)$$

The two terms expose separate failure modes: an inaccurate lateral estimate and a representation that changes too quickly for a one-step shift. The bound does not guarantee a useful shifted signal, so we report this variant only as an ablation.

2.3 Exact local gradients

Write $g = [\gamma - v]_+$. For TEREL-S the activation gradient is

$$e_t = \frac{2\lambda_S I_t}{N_I D} (z_t - p_t) - \frac{2\lambda_V}{BD} g \odot (z_t - m) + \frac{\lambda_C}{BD} Aq_t. \quad (8)$$

The input $z_t^{\ell-1}$ to each learned layer is detached. Consequently,

$$\frac{\partial L^\ell}{\partial W^\ell} = \sum_t (e_t \odot \phi'(a_t^\ell)) (z_t^{\ell-1})^\top, \quad \frac{\partial L^\ell}{\partial c^\ell} = \sum_t e_t \odot \phi'(a_t^\ell). \quad (9)$$

This is a presynaptic factor times an output-side teaching factor. Autograd computes ϕ' exactly; our derivation does not assume a linear activation. No later-layer error enters Eq. (9).

Canonical TEREL leaves adjacent $p_t = z_{t-1}$ inside the short chunk graph. Its temporal contribution is

$$\frac{2\lambda_S}{N_I D} [I_t(z_t - z_{t-1}) - I_{t+1}(z_{t+1} - z_t)], \quad (10)$$

with missing and boundary-crossing terms omitted. The second term depends on a future activation within the chunk. We therefore describe canonical TEREL as layer-local but not temporally online; TEREL-S is both layer-local and temporally local.

2.4 State update and execution order

After the encoder optimizer step, state is updated under `no_grad` from the activations computed for that step:

$$m^+ = \rho_m m + (1 - \rho_m) \bar{z}, \quad (11)$$

$$v^+ = \rho_m v + (1 - \rho_m) \frac{1}{B} \sum_t (z_t - m)^2, \quad (12)$$

$$A^+ = \rho_A A + (1 - \rho_A) \text{offdiag} \left(\frac{1}{B} \sum_t (z_t - m)(z_t - m)^\top \right). \quad (13)$$

The old mean m is used in Eqs. (12) and (13). For TEREL-S, the final activation and its centered value are stored as detached temporal state. No activation graph survives the call.

Algorithm 1 One TEREL-S step for a deep encoder.

Require: input batch/stream segment X , boundary flags b , layer states $\{m^\ell, v^\ell, A^\ell, p^\ell\}$

```

1:  $h \leftarrow X$ 
2: for layer  $\ell = 1, \dots, L$  do
3:    $z^\ell \leftarrow \phi(W^\ell \text{sg}(h) + c^\ell)$ 
4:   construct detached predecessors  $p_t^\ell$  and valid pairs from  $b$ 
5:   compute  $L^\ell$  by Eq. (5)
6:    $h \leftarrow \text{sg}(z^\ell)$ 
7: end for
8: backpropagate  $\sum_\ell L^\ell$  only within each layer; step optimizer
9: for layer  $\ell = 1, \dots, L$  do
10:  update detached state by Eqs. (11) to (13)
11: end for
```

2.5 Variants and claim boundary

Table 1: Variants used in the experiments. “Time graph” and “depth graph” indicate whether auto-grad crosses those boundaries.

Method	Temporal reference	Decorrelation	Graph scope
TEREL	live previous in chunk	tracked A , current q_t	same layer/chunk time
TEREL-S	detached previous	tracked A , current q_t	same layer/current step
Shifted proxy	detached previous	tracked A , previous q_t	same layer/current step
Direct control	detached previous	exact in-batch $\ C\ _F^2$	same layer/batch statistics
Local SupCon	label positives	supervised contrastive	same layer/current batch

Our primary locality claim is spatial: parameters receive no gradient from another layer. Canonical TEREL keeps short-range temporal credit within a chunk; TEREL-S additionally cuts credit across time and is online in the bounded-history sense when executed at batch size one. We do not equate these properties with biological plausibility or measured hardware efficiency.

2.6 State and memory cost

For layer width D , dynamic TEREL state contains mean D , variance D , lateral operator D^2 , previous activation D , previous centered activation D , and one validity flag: $D^2 + 4D + 1$ elements. A bounded streaming normalizer, when used, adds location and scale vectors of size $2D$. Both totals are constant in stream length and quadratic in width. Encoder parameters, optimizer moments, batch activations, and temporary matrix products are separate costs. In particular, AdamW adds approximately two parameter-sized moment buffers after its first step. The lateral state is discarded at inference, but its training cost is real. Batch-size-one execution removes batch growth and temporal graphs; it does not remove the dense lateral matrix.

Table 2: Conceptual comparison. Costs omit learned feedforward parameters.

Method	Temporal objective	Execution	Learned mapping	Auxiliary state
Batch SFA	derivative minimization	batch eigensystem	linear	covariance and whitening
IncSFA	derivative minimization	sample based	linear; deep cascades possible	incremental PCA and MCA state
Bio-SFA	derivative minimization	online neural dynamics	linear network in reported derivation	synaptic/dual variables
Blockwise SSL	paired-view invariance	mini-batch blocks	deep nonlinear blocks	batch statistics
TEREL family	temporal soft-SFA	sample or minibatch	deep nonlinear layers	$D^2 + 4D + 1$ per layer

3 Relation to prior work

SFA and incremental SFA. SFA is the objective-level ancestor of TEREL: temporal slowness, variance, and decorrelation are its defining ingredients [1]. Batch linear SFA whitens observations and solves a minor-component problem for their derivatives. IncSFA replaces those batch operations with covariance-free incremental PCA and minor-component updates, supports episodic boundaries, and has Hebbian/anti-Hebbian update forms [4]. It is therefore a stronger online precedent than a comparison to batch SFA alone. Our empirical baseline is a provenance-documented Python 3 port of the authors’ published linear algorithm, validated against batch SFA on a synthetic stream. TEREL differs by applying a soft objective directly to each nonlinear learned layer, at the cost of a dense tracked lateral operator.

Bio-SFA and trace learning. Bio-SFA derives an online neural network for the SFA problem with local synaptic updates [5]. This prevents us from presenting local online SFA as absent from prior work. TEREL instead contributes a different deep soft-penalty parameterization and an explicit stop-gradient implementation that is easy to compose with standard nonlinear layers. Földiák-style trace learning also exploits temporal persistence [6]; its positive activation trace is not algebraically the same as the signed difference in Eq. (8).

Regularized and blockwise self-supervision. VICReg and Barlow Twins use variance/covariance or cross-correlation penalties to prevent collapse without negatives [2, 3]. Blockwise self-supervised learning has already shown that VICReg, Barlow Twins, and contrastive losses can train network blocks without end-to-end backpropagation [7]. TEREL is therefore not distinguished by “a local VICReg loss” alone. Its narrower focus is temporal positives, detached streaming moments, and a same-layer covariance-gradient proxy that can execute one observation at a time.

4 Experimental protocol

The experiments answer four questions. First, after restoring the intended training schedule and optimizer state, is canonical TEREL competitive with compute-matched backpropagation and better than its untrained representation? Second, is that result carried only by early layers or retained by the final layer? Third, can TEREL-S learn non-collapsed representations one sample at a time with bounded state? Fourth, as a secondary stress test, does the temporal-coherence advantage transfer to a natural wearable stream?

4.1 Data roles and leakage control

MNIST controlled temporal supervision. The official 60,000-example training split is stratified with seed 1701 into 50,000 training and 10,000 validation examples. The official 10,000-example test split is excluded from v2 configuration recovery. Within each encoder epoch, training examples are randomly arranged into same-class chunks of length 16; all chunk starts are temporal boundaries. Labels create the persistent order but do not enter the TEREL loss. This controlled setting isolates whether temporal continuity can provide a useful local teaching signal; it is not presented as unlabeled self-supervision.

PAMAP2 secondary natural-order stress test. PAMAP2 contains 100-Hz wearable measurements and activity annotations from nine subjects [8]. We use its 12 protocol activities, discard transition label 0 as instructed by the dataset documentation, and downsample by a fixed stride of 10. Missing sensors are imputed with training-subject means and standardized using training-subject statistics. Subjects 1–6 train the encoder and probe, subject 7 is validation, and subject 8 is the untouched test subject. Subject 9 is excluded before model selection: its protocol recording contains only rope jumping and cannot support a 12-class test. Boundaries are inserted at subject starts, removed-label gaps, and timestamp discontinuities. The encoder iterator returns only features and boundaries; activity labels exist only in downstream evaluation. This experiment tests generalization of the temporal intuition; it does not replace the controlled MNIST comparison.

4.2 Models, baselines, and readouts

The primary encoder is $784 \rightarrow 512 \rightarrow 256$ with LeakyReLU (negative slope 0.01). Canonical TEREL uses batch normalization, live temporal pairs within chunks, and greedy layerwise training for 30 epochs per layer. Thus both its encoder and the matched BP encoder receive 60 complete training-set presentations. Its AdamW learning rate is 3×10^{-4} , weight decay 10^{-4} , batch size 256, and coefficients $(\lambda_S, \lambda_V, \lambda_C) = (1, 2.5, 1)$. Population momentum is 0.9; the lateral estimate uses the current batch ($\rho_A = 0$).

Backpropagation uses the same hidden dimensions, activation, normalization, optimizer, batch size, and 60 data passes, trained end to end with cross entropy. TEREL-S uses detached temporal references, joint local training, no batch normalization, $(1, 5, 1)$, and a separately validated ten-pass schedule. It is an intentionally lean temporally local variant, so its compute is reported rather than described as matched to BP. Random features retain the same seed-specific hidden weights without encoder updates.

The primary readout for every method is the same linear classifier over the concatenation of both hidden layers. A final-layer-only probe on the identical canonical encoder is a secondary audit of depth specialization. All probes use AdamW for 60 epochs (learning rate 0.003, weight decay 10^{-4} , batch size 2,048). This matched-access design prevents a comparison in which a local method exposes less of its representation than the baseline.

4.3 Validation recovery and freeze

Selection uses only the 10,000-example validation split and seeds 101, 202, and 303. We first reconstructed the original successful ingredients and then varied one factor at a time: temporal detachment, joint versus greedy layer training, lateral-state timescale, normalization, readout, augmentation, and schedule. The principal regression was not caused by the corrected loss: the failed protocol had simultaneously shortened training from 60 to 10 passes, changed greedy to joint optimization, detached time, removed normalization, restricted the readout, and set $\rho_A = 0.99$ in place of current covariance.

Table 3: Corrected MNIST test accuracy (% , five seeds). All-layer readout access is matched across methods; TeReL last-layer access is a secondary audit. Values are mean $\pm$ sample SD.

Method	Accuracy
TEREL (all layers)	97.30 $\pm$ 0.07
TEREL (last layer)	97.14 $\pm$ 0.10
TEREL-S (all layers)	96.29 $\pm$ 0.07
Random features (all layers)	95.13 $\pm$ 0.27
Backpropagation (all layers)	98.34 $\pm$ 0.08

Table 4: Paired test-set accuracy differences in percentage points with percentile 95% bootstrap intervals over seeds.

Contrast	Difference [95% interval]
TEREL – BP	-1.04 [-1.07, -1.01]
TEREL – random	2.17 [2.00, 2.42]
last – all layers	-0.16 [-0.30, -0.02]

A bounded validation-only recovery restored those ingredients before the confirmatory matrix was frozen.

The streaming configuration is selected separately because batch-size-one execution changes both normalization and update count. It uses TEREL-S, bounded running normalization, two passes, learning rate 6×10^{-6} , statistics and lateral momenta 0.99, and one forward/backward step per example. No future example, retained temporal graph, or stream-length-dependent buffer is used.

4.4 Confirmatory analysis and test gate

Central results use seeds 1101, 1202, 1303, 1404, and 1505. We report every raw result, the mean, sample standard deviation, and a percentile 95% bootstrap interval of the mean from 10,000 resamples (analysis seed 260803). Primary contrasts pair methods by seed and bootstrap the five paired differences. The preregistered competitive gate places canonical TEREL within 1.5 percentage points of BP; the learned representation gate requires improvement over matched random features. Non-collapse requires every canonical seed to have effective rank at least 20, median variance at least 0.05, and active fraction at least 0.25.

Before the v2 test run, all implementation tests passed; the validation ledger, resolved matrix, code commit, and protocol were hash-locked in a clean manifest; and the runner required an explicit test flag. The MNIST test set was not used during v2 recovery, but it had been observed by the rejected submission and by the earlier diagnostic revision. We therefore treat v2 as a sequential confirmation rather than calling the test split historically untouched. No v2 hyperparameter was changed after opening the gate. The subsequent mechanism audit was separately frozen and uses validation data only; it removes one loss coefficient or shuffles order without selecting a replacement model.

5 Results

5.1 Competitive corrected performance

Canonical TEREL reaches $97.30 \pm 0.07\%$, compared with $98.34 \pm 0.08\%$ for matched BP. The paired gap is 1.04 percentage points (95% bootstrap interval [1.01, 1.07]), inside the frozen 1.5-point competitive

gate. Relative to the identical untrained representation, TEREL gains 2.17 points ($[2.00, 2.42]$). All five seeds satisfy the non-collapse criteria, and every intended layer and normalization group has a nonzero parameter update. The corrected evidence therefore supports the paper’s central empirical statement: a deep network trained by layer-local soft-SFA reaches near-backprop MNIST accuracy.

5.2 Mechanism isolation

After the confirmatory matrix was complete, we froze a validation-only one-factor audit under the recovered canonical protocol. Removing the temporal term loses 8.18 percentage points without collapsing rank, while fully shuffling the class-persistent order loses 7.55 points. Temporal coherence is therefore the task-aligned teaching signal rather than an incidental regularizer. The two remaining terms prevent different degeneracies: without gated variance expansion, median feature variance falls to 7.7×10^{-5} ; without decorrelation, variance remains high but effective rank falls to 2.8. Batch normalization alone does not rescue either failure. The three signals thus have empirically distinct roles matching the soft-SFA interpretation.

Table 5: Validation-only mechanism audit under the recovered canonical protocol (three paired seeds). Each intervention changes one factor. Δ is the accuracy difference from full TEREL in percentage points.

Intervention	Accuracy (%)	Δ	r_{eff}	Median var.
Full TEREL	97.28 ± 0.12	–	39.2	0.185
$\lambda_S = 0$	89.10 ± 1.57	-8.18	369.3	0.907
$\lambda_V = 0$	66.13 ± 6.32	-31.15	6.0	7.72e-05
$\lambda_C = 0$	86.44 ± 0.41	-10.84	2.8	0.916
Shuffled order	89.73 ± 0.37	-7.55	311.3	0.171

5.3 Depth, readout, and representation geometry

The last hidden layer retains almost all of the learned signal: its probe reaches $97.14 \pm 0.10\%$, only 0.16 points below the matched all-layer readout ($[-0.30, -0.02]$). Thus competitive performance is not an artifact of concatenating a strong first layer while ignoring a weak second layer. TEREL-S reaches $96.29 \pm 0.07\%$ after only ten joint-training passes and beats random features by 1.16 points ($[0.94, 1.45]$), showing that temporal detachment trades some accuracy for a stricter locality boundary rather than destroying the mechanism.

The diagnostic geometry agrees with the probe result. Canonical TEREL raises nearest-centroid accuracy from 80.21% for random features to 95.55% and the between/within-class scatter ratio from 0.24 to 0.70. Its final layer is even more class concentrated (96.22% centroid accuracy and 1.52 scatter ratio), while the all-layer representation keeps greater effective rank. This is the expected soft-SFA tradeoff: persistent factors become accessible without forcing every output direction into the same class-compressed geometry.

Table 6: MNIST representation diagnostics averaged across five frozen seeds. B/W is between/within-class scatter, Sel. is median unit class selectivity, r_{eff} is covariance effective rank, and $|\rho_{\text{off}}|$ is mean absolute off-diagonal correlation.

Method	Centroid (%)	B/W	Sel.	r_{eff}	$ \rho_{\text{off}} $
Random (all)	80.21	0.241	1.239	33.84	0.101
TeReL-S (all)	92.04	0.101	0.843	142.32	0.057
TeReL (all)	95.55	0.695	1.704	37.35	0.103
TeReL (last)	96.22	1.516	2.567	18.86	0.161
BP (all)	97.87	0.968	1.857	21.17	0.152

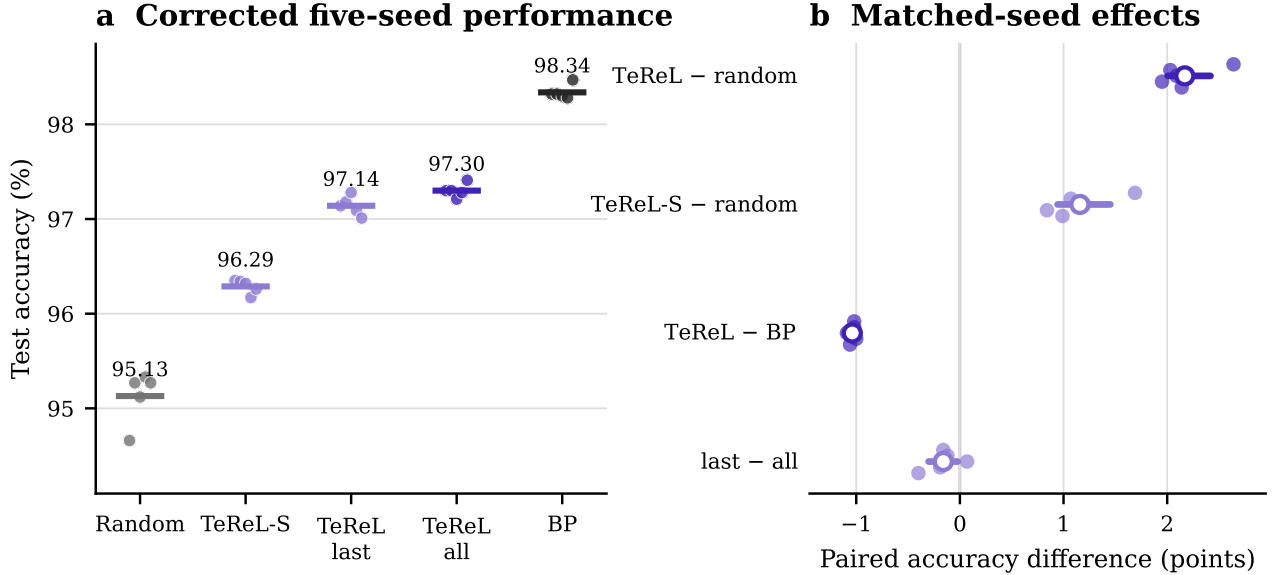


Figure 2: Corrected confirmatory evidence. **a**: individual MNIST test seeds and means (horizontal bars). Canonical TeReL recovers near-BP accuracy, its final layer retains the result, and temporally local TeReL-S remains well above random features. **b**: paired seed differences; lines are percentile 95% bootstrap intervals and open circles are means.

5.4 Natural temporal order on PAMAP2

The chronological/shuffled PAMAP2 contrast was introduced to answer whether the controlled temporal mechanism transfers when order is natural and labels are excluded from encoder batches. Architecture, initialization, optimizer, sample count, and probe are held fixed; only encoder order changes. The result is inconclusive: chronological and shuffled learning reach 0.307 and 0.312 macro-F1, respectively, with a paired interval spanning both directions ($[-0.063, 0.053]$). All methods also exhibit a large subject-7 to subject-8 shift. We report this boundary because it is informative, but do not treat a single heterogeneous wearable split as a reversal of the controlled MNIST result or of the temporal-coherence motivation. Broader natural-stream tests need multi-subject validation and stronger shift control.

5.5 Streaming execution and cost

The first untuned batch-size-one audit verified bounded state but produced weak rank. We treated that as a starting point rather than a conclusion and tested learning rate, state timescale, streaming normalization, accumulation, and exposure count on validation data. The selected two-pass configuration reaches $95.14 \pm 0.04\%$ accuracy across three seeds, with effective rank 112.8 (seed range 106.6–120.9) and median feature variance 0.378. It exceeds the matched random-feature validation mean by 0.29 points while retaining no temporal graph.

At every step, the previous activation, running moments, normalization state, and lateral matrix are detached; state size is unchanged by stream length. The serial resource seed processes one observation per optimizer step and takes 1,305 seconds, illustrating that a stock GPU/PyTorch implementation pays a large launch and update overhead. This is evidence for quality-bearing samplewise learning and bounded storage, not a claim that the present software is throughput- or energy-optimal. Encoder resource accounting for the main methods is given in [Tables 9](#) and [10](#).

6 Limitations

TEREL inherits the temporal-coherence assumption: rapidly changing or aliased latent causes can make adjacent observations misleading. The MNIST ordering is synthetic and label-derived, so its high accuracy establishes the efficacy of the local temporal teaching signal rather than label-free learning. PAMAP2 is a modest, distribution-shifted sensor benchmark, and its single subject-disjoint split cannot decide the broader natural-order question. Scaling to images, video, audio, and additional natural streams remains an empirical requirement.

The dense lateral operator gives bounded history but $O(D^2)$ state and within-layer communication. Sparse or low-rank approximations need direct quality tests. The streaming result is validation evidence on MNIST, not a five-seed test-set or throughput-matched comparison; samplewise optimizer steps are slow in the present GPU implementation. Canonical TEREL also takes more wall-clock time than BP despite a lower declared operation proxy, so no hardware or energy advantage follows from these software measurements.

The five-seed v2 matrix was frozen after validation-only recovery, but the official MNIST test set was historically seen by the rejected submission and the separate diagnostic revision. This is a sequential confirmation, not a pristine first test access. An independent benchmark should provide the next strong confirmation. Finally, locality here concerns credit assignment: standard optimizers, exact activation derivatives, normalization, and synchronized matrix operations remain.

7 Perspective and future directions

Natural order. SFA’s motivating regime is not class-sorted images but streams in which stable causes generate many changing observations. The PAMAP2 result shows that adjacency alone is insufficient under severe subject shift; it does not remove the underlying opportunity. Video under controlled viewpoint change, egocentric interaction, and continuous audio provide cleaner tests in which the timescale of the latent cause can be compared directly with the TEREL state and chunk timescales. A useful next study should vary those timescales rather than reduce natural order to one chronological-versus-shuffled number.

Constrained hardware. The rule’s ingredients—presynaptic activity, an output-side teaching factor, leaky state, and no error path across depth—are structurally compatible with layer-local or event-driven execution. TEREL-S further removes the temporal graph, and our samplewise experiment

shows that this stricter form can retain useful accuracy. These are architectural reasons to investigate specialized implementations, not measurements of energy efficiency. Low-rank lateral state, fused state updates, and optimizer choices designed for local arrays are the concrete engineering steps needed to test that perspective.

Recurrent extension. The temporal objective also suggests a recurrent representation module whose slow state supports prediction without receiving prediction gradients. The earlier exploratory model did not enforce that separation and therefore cannot test the idea cleanly. A valid recurrent comparison should isolate the representation and prediction paths, include matched random-state and BPTT controls, and repeat across seeds. We retain this as a motivated extension of the mechanism, not as a result established here.

8 Conclusion

Temporal coherence, variance preservation, and decorrelation form a useful representation-learning recipe that connects modern regularized self-supervision to SFA. TEREL turns that recipe into a concrete deep local rule: each nonlinear layer receives a temporal error, a gated neuron-local expansion signal, and a same-layer decorrelation signal. Canonical TEREL keeps short within-chunk temporal credit while cutting the error path across depth. TEREL-S additionally detaches time and has stream-length-independent state, although its dense lateral operator is quadratic in width.

After restoring the intended optimizer, schedule, normalization, and lateral timescale, canonical TEREL reaches 97.30% on MNIST, within 1.04 points of compute-matched BP and 2.17 points above matched random features. The final layer alone retains 97.14%, while temporally local TEREL-S reaches 96.29% and its tuned samplewise form learns a high-rank 95.14% validation representation. One-factor controls further show that temporal coherence supplies the aligned signal, variance expansion prevents scale collapse, and decorrelation prevents redundancy collapse. These results substantiate the paper’s central intuition: temporal coherence, protected against collapse and redundancy, can train a competitive deep representation with local credit assignment. The secondary natural-stream result and explicit cost accounting mark the next experiments without displacing that result.

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A Gradient derivations

For valid detached pairs,

$$\frac{\partial}{\partial z_t} \frac{1}{N_I D} \sum_s I_s \|z_s - \text{sg}(z_{s-1})\|^2 = \frac{2I_t}{N_I D} (z_t - p_t). \quad (14)$$

Because g , m , A , and q_t are detached,

$$\frac{\partial L_V}{\partial z_t} = -\frac{2}{BD} g \odot (z_t - m), \quad (15)$$

$$\frac{\partial L_C}{\partial z_t} = \frac{1}{BD} A q_t. \quad (16)$$

Combining terms gives Eq. (8), and applying the chain rule to $z_t = \phi(Wx_t + c)$ gives Eq. (9). For undetached adjacent pairs, z_t appears as the current member of pair t and the previous member of pair $t + 1$; summing those derivatives gives Eq. (10). Finite-fixture regression tests check both signs and the presence of the future term.

For the direct covariance control, let Z_c stack centered rows and $C = \text{offdiag}(Z_c^\top Z_c / B)$. Since C is symmetric,

$$\frac{\partial}{\partial Z_c} \frac{\|C\|_F^2}{D} = \frac{4}{BD} Z_c C, \quad (17)$$

equivalently $4C(z_t - m)/(BD)$ in column-vector notation. The proxy replaces $4C(z_t - m)$ by the detached direction Aq_t and absorbs the positive scale into λ_C .

B Implementation-fidelity tests

The public suite checks:

- the negative variance gradient sign, variance gate, temporal detachment, undetached future derivative, lateral-proxy error bound, and direct covariance fixture;
- nonzero updates for every declared encoder layer and lateral state, with no gradient from a later layer to an earlier layer;
- fixed dynamic-state size, bounded streaming normalization, samplewise gradient release, and absence of retained autograd graphs;
- exact MNIST split roles, PAMAP2 subject roles, train-only normalization, temporal boundaries, and label-free natural-stream batches;
- matched all-layer and last-layer probes, greedy layerwise scheduling, execution of every declared BP layer, and exact data-pass accounting;
- batch SFA constraints, boundary-aware derivatives, and numerical validation of the official-algorithm IncSFA port; and
- refusal of held-out evaluation after protocol, ledger, commit, or tracked-worktree drift.

C Reproducibility record

The repository pins Python 3.12 and all direct dependencies in `pyproject.toml` and `uv.lock`. Every run enables deterministic PyTorch algorithms, deterministic cuDNN behavior, and the prescribed cuBLAS workspace configuration. The protocol, selection plan, validation ledger, confirmatory manifest, per-seed JSON records, raw confusion matrices, hardware/software record, and generated paper tables are checksum linked. The executed identifiers are:

```
code          02afd90cf6927a588aa424d61cb86c6876b25c17
manifest      9a1f8b8e5b75293efcbe5b13a767e74f57370f91658c73deb51e808e6c1c9828
config        3e83ea66c03a621bbdc6f1a16a143c6dbf50643ae2b5213a1d78afba68ed0a6b
protocol      faab529f7dab0d2940a8e7f6dd662c482b9da64bcf661e78029378c9fb0a68da
validation    11faf7a030f9f76fd2174225ca8ba806d9439e0bbf63924b01361909ee3305cb
results       727a0e9c335d5281cd3037ef8f64ee68ba927e9e77e4c0741312d0e4a7dcde88
analysis      e5867843aceb97bb27b9390902faf8bbe159df9d101b4bb2c4f51566dd722d30
mcode         9532b3476c36e981dd5164c2c43affd4e791c9c6
mconfig       c061c4f761e2037265a2c2b45033d05bc11c62c3470b6a3390756a6067c80d28
mprotocol     505db43ba8cb06d5175beb163f0828c58b58afcb9b66a95217e15b8abf1cd41e
mrecords      53c67637a43b4f92227db1a19a144a049b934f34ca1ac5506223b4971fcee6bc
manalysis     9f7378302e47219ef14e0d7a30d20a8f37417a5595ae8a94e502944b1bf43d61
```

The core commands are:

```
uv sync --extra test
uv run pytest -q
uv run python -m terel.resubmission.confirmatory run \
  --manifest <confirmatory-manifest-v2.json> \
  --protocol <confirmatory-protocol-v2.md> \
  --validation-ledger configs/resubmission/validation-ledger-v2.json \
  --repository . --allow-test --output <directory> --device cuda
uv run python -m terel.resubmission.analysis_v2 \
  --results <confirmatory-results-v2> \
  --streaming-results <streaming-recovery-v2> \
  --analysis-output <analysis-v2.json> \
  --results-tex <results-v2.tex> --appendix-tex <appendix-v2.tex>
uv run python -m terel.resubmission.mechanism_analysis_v2 \
  --reference-results <recovery-v2> --audit-results <mechanism-v2> \
  --analysis-output <mechanism-analysis.json> \
  --results-tex <generated-mechanism.tex>
```

Absolute data and artifact roots in the executed plan are retained for provenance; the README gives portable equivalents. Dataset files are not redistributed.

D Secondary natural-stream controls

The PAMAP2 stress test was executed in the corrected v1 matrix before the v2 performance recovery. Its encoder never receives activity labels. The chronological and shuffled TEREL rows differ only in order; the remaining rows locate their performance relative to untrained features, classical SFA, the exact in-batch covariance control, and end-to-end supervision. All methods are weak on subject 8, confirming that the split is dominated by cross-subject shift. The immutable identifiers are:

pamap code abfa0ec1df78204b66d0c34141f5deb6063a572a
 pamap manifest 38ca928e0884a8bc40a5c7ffefc46313fc96c170661aea63f93bf9496b2b1af0
 pamap results ad817a220fae65f9ebceb5941f967ec5bff2015c10847d1f8b00f54032ce7213

Table 7: Secondary PAMAP2 test macro-F1 across five seeds. Values are mean $\pm$ sample SD and percentile 95% bootstrap interval.

Method	Macro-F1
Chronological TEREL	0.307 ± 0.066 [0.260, 0.360]
Shuffled TEREL	0.312 ± 0.026 [0.290, 0.330]
Random features	0.322 ± 0.039 [0.296, 0.355]
Batch SFA	0.185 ± 0.002 [0.184, 0.186]
Incremental SFA	0.218 ± 0.015 [0.206, 0.229]
Direct covariance	0.243 ± 0.021 [0.226, 0.259]
Supervised BP	0.319 ± 0.030 [0.296, 0.341]

E Complete configurations and raw results

Table 8: Raw corrected MNIST test accuracies (%) in seed order 1101, 1202, 1303, 1404, 1505.

Method	Raw values
TEREL (all layers)	97.30, 97.30, 97.21, 97.28, 97.41
TEREL (last layer)	97.14, 97.18, 97.28, 97.09, 97.01
TEREL-S (all layers)	96.35, 96.34, 96.32, 96.17, 96.26
Random features (all layers)	94.66, 95.27, 95.12, 95.33, 95.27
Backpropagation (all layers)	98.32, 98.32, 98.30, 98.28, 98.47

Table 9: Encoder-fit accounting averaged over five test seeds. Examples count sample presentations. MAC is the declared linear plus same-layer proxy, not a hardware energy measurement.

Method	Examples (M)	Opt. steps	Seconds	Peak MiB	MAC (G)
TEREL (all layers)	3.00	11760	107.6	76.7	3981.3
TEREL (last layer)	3.00	11760	107.7	76.7	3981.3
TEREL-S (all layers)	0.50	1960	27.4	77.7	1126.4
Backpropagation (all layers)	3.00	11760	43.4	225.4	4815.4

Table 10: Tuned samplewise streaming TeReL on the validation split (three seeds). The serial seed supplies time and memory accounting; concurrent repetition timings are excluded.

Accuracy (%)	Effective rank	Median variance	Encoder seconds
95.14 ± 0.04	112.8	0.378	1305.2